

IB 514: Scientific Writing

Mark Novak

Dept. of Integrative Biology, Oregon State University

Contents

1	Introduction	3
1.1	Philosophy	3
1.1.1	Course Goals	3
1.1.2	What you'll need	3
1.1.3	Core Principles	3
1.2	Overview	3
1.2.1	Website & Materials	3
1.2.2	Questions?	3
1.2.3	Who are we?	3
1.2.4	Accountabilibuddies	3
1.3	My hope	3
2	Habits	3
2.1	Daily Writing Routines	3
2.2	Time Management & Deadlines	3
2.3	Early Drafting Practices	3
2.4	Writer's Block	3
2.5	Meta-Cognition	3
2.6	Assignment	3
3	Workflows	3
3.1	Project Structure and Organization	3
3.2	Writing Workflows	3
3.3	Writing Tools	3
3.4	Figures	3
3.5	Final thoughts	3
4	Metastructure	3
4.1	Audience & Journal	3
4.2	Organization and Flow	3
4.3	Journal Article Types	3

4.4	Content Strategies	3
4.5	Additional Sand-Jensen Principles	3
4.6	Final Thoughts	3
5	Paperanalysis	3
5.1	Individual summary	3
5.2	Small-group discussion I	3
5.3	Small-group discussion II	3
5.4	Takeaways	3
5.5	Implementation	3
6	Rainbowparagraphs	3
7	Unfilteredflow	3
7.1	Warm-Up	3
7.1.1	Zero Filter	3
7.1.2	Stream of Consciousness	3
7.2	Clarity and Velocity	3
7.2.1	Jargon Free	3
7.2.2	Volume Competition	3
7.3	Framing and Structure	3
7.3.1	Reverse Logic	3
7.3.2	Methods in Motion	3
7.3.3	Figure Narrative	3
7.3.4	Abstract Compression	3
7.4	Overall Debrief	3
8	Reverseoutline	3
9	Literature	3
9.1	Framing the Challenge	3
9.2	Staying Current	3
9.2.1	Alerts and Feeds	3
9.2.2	Active Filtering	3
9.3	Literature Search	3
9.3.1	Search Strategies	3
9.3.2	Search Tools	3
9.4	Reference Management	3
9.4.1	Tools	3
9.4.2	Organisation and Annotation	3
9.5	Writing from the Literature	3
9.5.1	Integration Strategies	3
9.5.2	Hybrid Writing Strategy	3
9.6	Takeaway	3
10	Collaboration	3
10.1	Guiding Principles	3

10.2 Tools and Conventions	3
10.3 Collaboration is Communication	3
10.4 Navigating Disagreements	3
10.5 Task Management	3
10.6 Authorship Conversations	3
10.7 Hierarchy and Power Imbalance	3
10.8 Final Thoughts	3
11Peerreview	3
11.1 Pipeline	3
11.2 Editor’s Perspective	3
11.3 Reviewer’s Perspective	3
11.4 Author’s Perspective	3
11.4.1 First steps	3
11.4.2 Rejection Management	3
11.5 Revising & Resubmitting	3
11.5.1 Workflow	3
11.5.2 Response Letter	3
11.6 Proofing Process	3
11.7 Peer Review History	3
12Peerintroanalysis	3
12.1 Format	3
12.2 Feedback Questions	3
12.3 Debrief	3
13Journalprep	3
13.1 Final Formatting	3
13.2 Author Contributions	3
13.3 Page Charges	3
13.4 Friendly Reviewers	3
13.5 Suggesting Reviewers	3
13.6 Submission Logistics	3
13.7 Open Science	3
13.8 Preprint Repositories	3

1. Introduction

1.1 Philosophy

1.1.1 Course Goals

1.1.2 What you'll need

1.1.3 Core Principles

1.2 Overview

1.2.1 Website & Materials

1.2.2 Questions?

1.2.3 Who are we?

1.2.4 Accountabilibuddies

1.3 My hope

2. Habits

2.1 Daily Writing Routines

2.2 Time Management & Deadlines

2.3 Early Drafting Practices

2.4 Writer's Block

2.5 Meta-Cognition

2.6 Assignment

3. Workflows

3.1 Project Structure and Organization

3.2 Writing Workflows

3.3 Writing Tools

3.4 Figures

3.5 Final thoughts

4. Metastructure

4.1 Audience & Journal

4.2 Organization and Flow

4.3 Journal Article Types

4.4 Content Strategies

4.5 Additional Sand-Jensen Principles

4.6 Final Thoughts